

# John Maliha | CPI en génie informatique

Montréal, QC

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## Profile

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Innovative and detail-oriented **Computer Engineer** with expertise in **embedded systems, algorithmic optimization, and full-stack development**. Hands-on experience in **hardware and software design**, with a passion for solving real-world problems in interdisciplinary fields such as **biomedical engineering, automation, and artificial intelligence**.

## Professional Experience

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**GIRO** **Montreal**  
*Solution Developer (Software Engineer – Optimization / Développement)* *2024 – Present*

- Developed **C++ optimization algorithms** for clients' planning, improving scheduling efficiency by **15%**.
- Designed and implemented new C++ features for international clients (Golden Gate, NYC, Los Angeles, Oakland).
- Automated testing workflows with **Google Test, Silk**, and Jenkins pipelines, reducing QA time by **20%**.
- Specialized in **mapping systems**: importing, creating, and customizing geospatial data for transit organizations.
- Delivered technical training and workshops for **5+ transit authorities** (MTAB, CTA, LACMATA, GCRTA, SFMTA).
- Automated workflows using Bash, PowerShell, and Python scripts, reducing manual processing time by **30%**.
- Managed SQL and Oracle databases for large-scale enterprise solutions.
- Collaborated in Agile environment using TFS and Git for version control.

**Mouvement Desjardins** **Montreal**  
*Software Developer / Computer Engineering Intern* *2022 – 2023*

- Migrated API from C# 6 to C# 10, boosting performance and improving security compliance.
- Built a **Python application** to automatically detect discrepancies in PDF documents, improving accuracy by **25%**.
- Implemented CI/CD pipelines with **Docker, Azure, and AWS**, reducing deployment time by **25%**.
- Optimized and refactored existing codebase for maintainability and scalability.
- Enhanced internal automation API (**Quadiant Automation**) by introducing major new features, adopted company-wide.
- Promoted from intern to **full-time developer** and later technical lead within one year.
- Developed a Windows application in C# for system log archiving and monitoring.
- Reduced code smells by 40% using **SonarQube**.
- Deployed and maintained software on both **DEV and PROD** environments.

**ASTP – Polytechnique Montréal** **Montreal**  
*VP of Technology (Technical Lead – Web Development)* *2022 – 2024*

- Maintained and upgraded the association's WordPress website serving **10+ student technical societies**.
- Led the design and development of a new platform using **JavaScript, Spring, HTML, and CSS**.
- Coordinated a small development team and provided technical mentorship.

## Projects

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**Polytechnique Montréal**  
*Drone Web Application (Academic Project)* *2022*

- Developed drone firmware (Crazy-fly) in C and simulated drone swarms in C++ (Argos).
- Built server in Python and web client in Angular/TypeScript for real-time drone monitoring.
- Debugged electronic systems during firmware development.
- Implemented and optimized pathfinding algorithms (**A\***, **Dijkstra**) for navigation.
- Designed interactive real-time map visualization for drone movement.

**Montreal**  
2023

#### *Lebanese Festival Platform (Volunteer Project)*

- Designed and developed cross-platform website (Angular + PHP) for festival coordination.
- Automated volunteer scheduling using Excel integration and email distribution.
- Secured user data and authentication with **JWT**.

**Polytechnique Montréal**  
2021

#### *Drawing Web Application (Academic Project)*

- Built collaborative drawing application for tablet and web using Express.js + Angular.
- Implemented a NoSQL database with **MongoDB** for scalable data storage.

**Montreal**  
2021

#### *Veterinary Clinic Platform (Project)*

- Developed a web/mobile platform with Angular and PostgreSQL.
- Optimized database and server to handle multiple simultaneous connections with no latency.

**Montreal**  
2024

#### *Cloud Computing Platform (Project)*

- Designed cloud platform in Python and Terraform with AWS instances.
- Implemented Cloud Design Patterns (Proxy, Gatekeeper, Trusted Host).
- Deployed Flask applications on AWS EC2 instances.
- Containerized Python applications using Docker.
- Managed MySQL databases in both standalone and cluster configurations.
- Built multi-threaded Python code for high-volume request handling.

## Technical Skills

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**Programming:** Proficient in Python, C++, C, and C# for enterprise and embedded applications; experienced in Java, TypeScript, JavaScript, SQL (PostgreSQL, Oracle), PHP, Firebase, XML, HL7.

**Frameworks:** .NET, Angular, React, D3, Scikit-learn, Apache, Spark, Express.js, REST APIs, Spring, OOP design.

**Tools:** Docker, Kubernetes, Azure, Git, Jira, SonarQube, JFrog, YAML, MS Visio, MS Office, MS Project.

**Domains:** Embedded systems, algorithmic optimization, CI/CD automation, data visualization, applied AI, biomedical engineering, medical informatics, bioinformatics, epidemiology.

## Education

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### **Polytechnique Montréal**

*Bachelor of Computer Engineering*

2019 – 2023

Specialization in Artificial Intelligence.

Relevant coursework: Embedded Systems, Machine Learning, Cloud Computing, Bioinformatics, Data Structures, Databases.

## Languages

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**French:** Native

**English:** Native